

★ WEEK 7 · PART 3 · PRACTICE

Error Correction and Pairs, by Hand

The Speed of Reversion, and the Trade It Powers

From the adjustment coefficient to a z-score trade

Part 3's payoff, with the arithmetic shown. We read the error-correction coefficient to find how fast a spread reverts and its half-life, then turn the spread into a z-score signal - entering, exiting, and stopping - to run the pairs trade the whole week built toward.

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PART 3 · APPLICATIONS

From Restoring Force to Trade

Case A: the speed of adjustment - how fast the gap closes - and the half-life of mean reversion. Case B: normalizing the spread into a z-score and turning it into entry, exit, and stop rules with a worked profit. Then a Claude Code lab.

What Part 3 Told Us

Now we trade it - with numbers

1

Error correction is the restoring force

When cointegrated series drift apart, the equilibrium error pulls them back. The adjustment coefficient measures how hard.

2

The half-life sets the clock

From the adjustment speed, the half-life tells you how many days until half the gap is gone - your expected holding period.

3

The z-score turns the spread into a trade

Normalize the spread, fade the extremes, and exit at the mean. Entry, exit, and stop rules follow directly.

Everything here is a calculator exercise: one adjustment speed, one half-life, one round-trip trade.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from the spread and its dynamics.
- Read the adjustment speed, or the z-score.
- Then act on the number.

What you should watch

- How a negative alpha closes the gap.
- How the half-life sets the holding period.
- How the z-score triggers the trade.

The Speed of Reversion

How fast the gap closes

When the spread strays from equilibrium, error correction pulls it back. The adjustment coefficient - alpha, negative - is the fraction of the gap erased each day. From it we get the half-life: how long until half the deviation is gone. That number is the expected holding period of a pairs trade, so it's worth computing exactly

The setup

- The spread from our cointegrated pair.
- Its error-correction coefficient is -0.25.
- It reverts toward equilibrium each day.

What we'll do by hand

- Read alpha as a fraction closed per day.
- Convert to the AR(1) coefficient.
- Compute the half-life.

The Error-Correction Model

A restoring force toward equilibrium

$$\Delta Y_t = \underbrace{\alpha (Y_{t-1} - \beta X_{t-1})}_{\text{pull back to equilibrium}} + (\text{short-run terms}) + \varepsilon_t \quad (\alpha < 0 \text{ is the restoring force})$$

The mechanism

- Last period's equilibrium error drives today.
- Alpha scales how strongly it pulls back.
- A negative alpha means a restoring force.

Who adjusts

- Each series has its own alpha.
- The larger one does more of the correcting.
- Sometimes only one side moves.

Read the Adjustment Speed

A fraction of the gap, per day

$$\alpha = -0.25 \Rightarrow \text{25\% of the gap closes each day,} \quad \phi = 1 + \alpha = 0.75 \text{ (spread AR(1))}$$

The sign and size

- Negative alpha pulls the spread back.
- Its magnitude is 0.25.
- So a quarter of the gap closes each day.

The AR(1) view

- The spread follows an AR(1) with $\phi = 0.75$.
- Each day it keeps 75% of its deviation.
- That sets up the half-life.

The Half-Life

How long until half the gap is gone

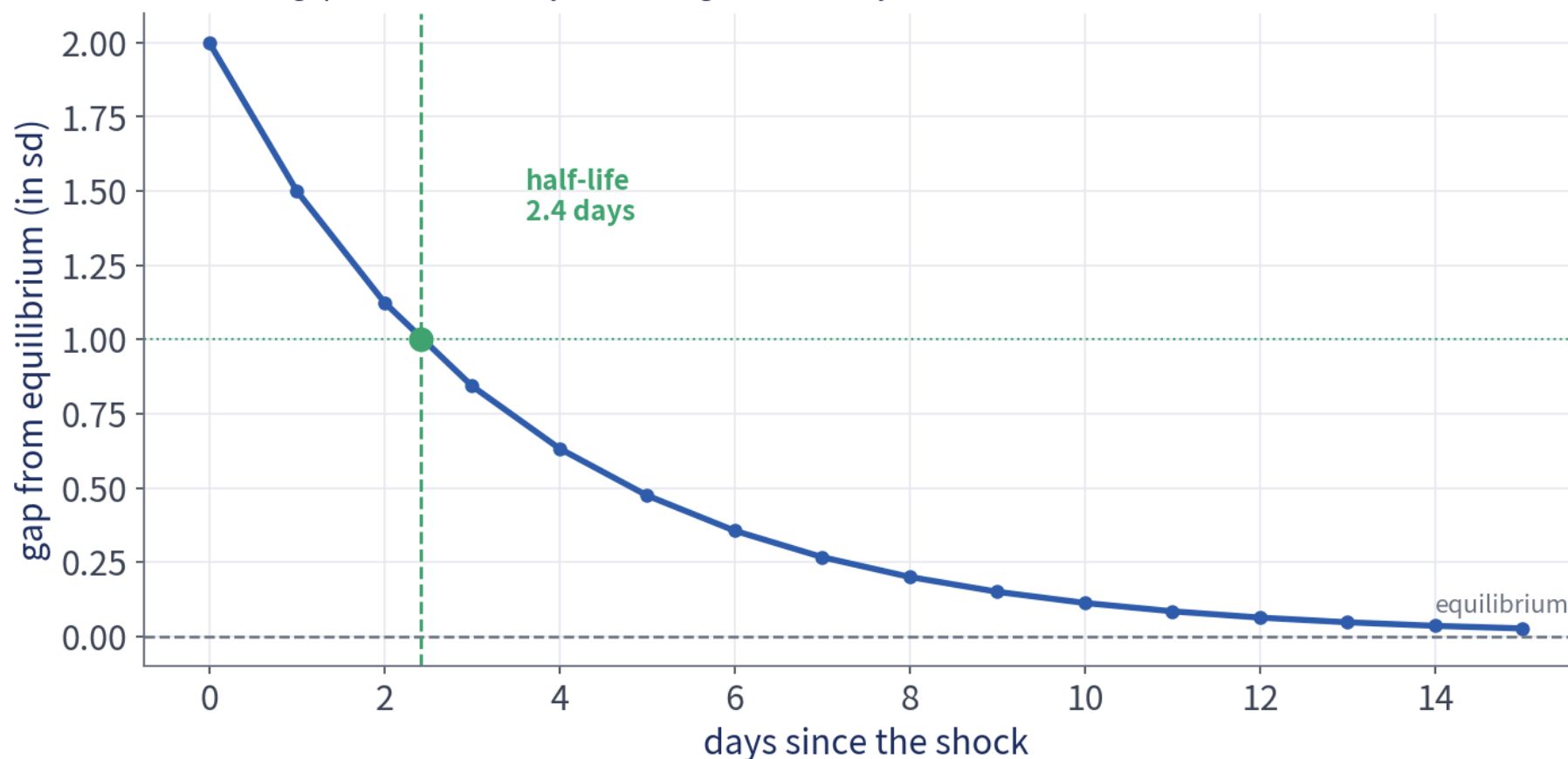
$$\text{half-life} = \frac{\ln 0.5}{\ln \phi} = \frac{\ln 0.5}{\ln 0.75} = \frac{-0.693}{-0.288} = \mathbf{2.4} \text{ days}$$

Dividing the log of one-half by the log of the AR(1) coefficient gives a half-life of 2.4 days. Half of any deviation is erased in under three days - a fast-reverting spread, ideal for a pairs trade.

The Gap Decays Away

Starting two standard deviations from equilibrium, the spread shrinks 25% a day, tracing a smooth exponential decay back to zero. It crosses the halfway line at 2.4 days - the half-life. The faster this decay, the shorter you hold each trade and the more often you can trade

Case A: the gap closes 25% a day - half of it gone in 2.4 days



The Half-Life Is the Clock

It sets the whole trade

What it tells you

- The expected holding period per trade.
- How many round trips a year you can make.
- Whether the edge is worth the capital.

What to avoid

- A near-zero alpha means glacial reversion.
- Long half-lives tie up capital for little.
- Prefer pairs that snap back quickly.

The Z-Score Trade

Normalize the spread, then act

With the hedge ratio and a fast half-life in hand, the trade is simple.

Standardize the spread into a z-score, so its units are standard deviations. When the z-score reaches plus or minus two, the spread is stretched - fade it. When it returns to zero, exit. If it pushes past three, the equilibrium may be breaking - cut the loss. We work a full

The spread stats

- Mean 5.2, standard deviation 0.62.
- The z-score rescales it in sigmas.
- Extremes are trade signals.

What we'll do by hand

- Find the entry and exit spread levels.
- Compute the captured spread.
- Set the stop.

The Z-Score Signal

Standard deviations from the mean

$$z_t = \frac{s_t - \bar{s}}{\sigma_s} = \frac{s_t - 5.2}{0.62} : \quad z = +2 \leftrightarrow s = 6.44, \quad z = 0 \leftrightarrow s = 5.20$$

Reading the z-score

- Zero means the spread is at equilibrium.
- Plus or minus two means it's stretched two sigma.
- The sign says which way to trade.

Turning it into levels

- A z of +2 is a spread of 6.44.
- A z of 0 is a spread of 5.20.
- Those are the entry and exit prices.

Entry, Exit, and Stop

Turning z into a trade plan

z-score	spread level	action
+2	6.44	ENTER: short 1 Y, long 1.96 X
0	5.20	EXIT: spread back at the mean
+3	7.06	STOP: equilibrium may be breaking

At a z of +2 the spread is stretched to 6.44 - short it. Exit when it reverts to 5.20 at z of zero. If instead it widens to 7.06 at z of +3, the pair may be decoupling: stop out.

The Round-Trip Profit

What the reversion pays

enter at $z = +2$ ($s=6.44$) : short 1 Y , long 1.96 X

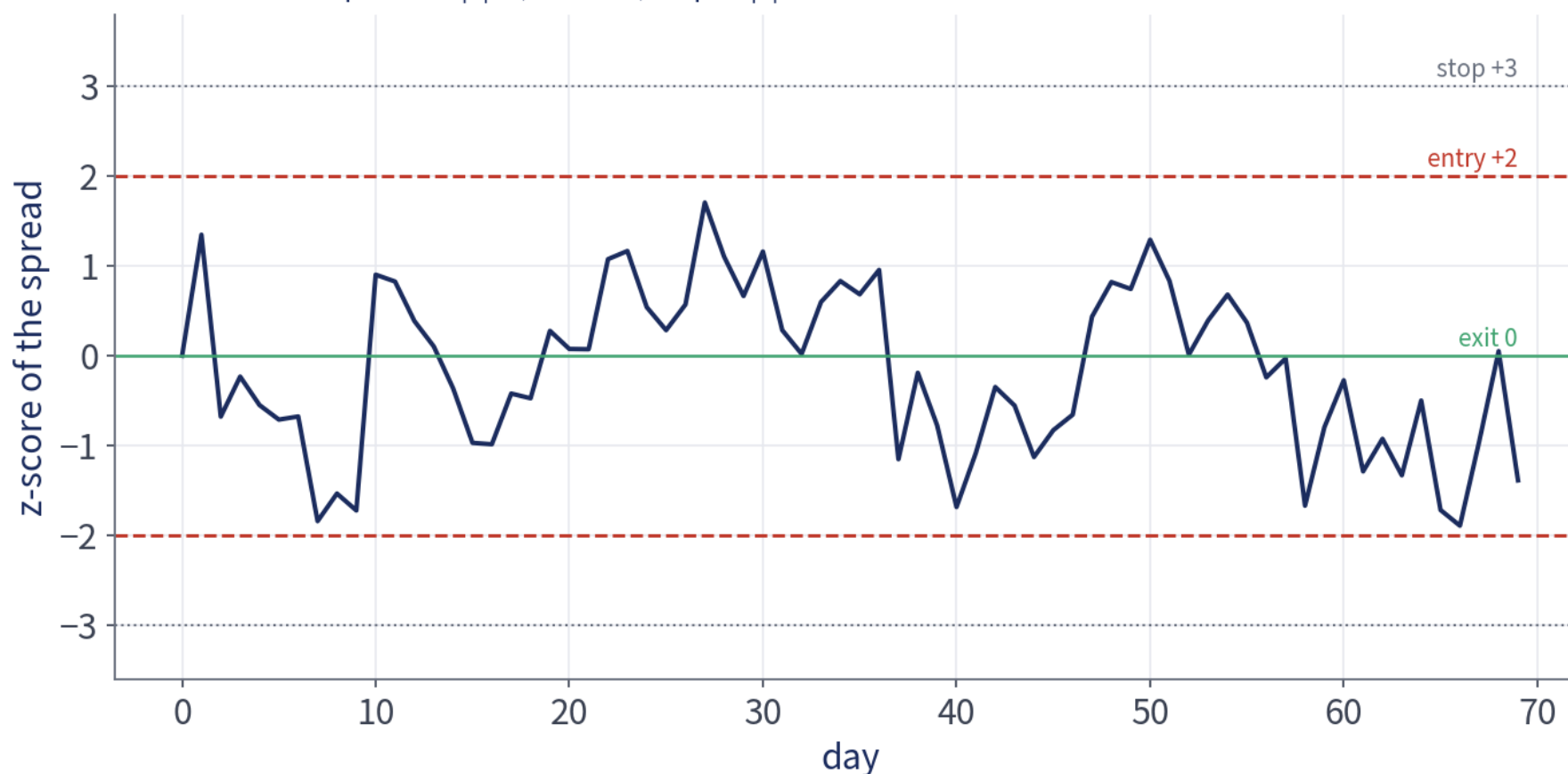
exit at $z=0$ ($s=5.20$) : captured $6.44 - 5.20 = \mathbf{1.24}$ per share of Y

Shorting the spread at 6.44 and covering at 5.20 captures 1.24 points per share of Y . With a 2.4-day half-life, the position reverts in about five days - a fast, repeatable round trip.

Fade the Extremes, Exit at Zero

The z-score oscillates around zero. Each time it pierces the plus-or-minus-two bands (red), the spread is stretched - enter against it. Exit as it returns to the green zero line. The dotted plus-or-minus-three lines are the stop: beyond them, the equilibrium may be breaking down

Case B: fade the spread at $|z|=2$, exit at 0, stop at $|z|=3$



The Trade, and Its Guardrail

Profit rests on the spread reverting

Why it works

- Cointegration guarantees the spread reverts.
- The z-score times entry and exit cleanly.
- The half-life bounds the holding period.

The guardrail

- Cointegration can break - watch for it.
- A z past three is a warning, not an opportunity.
- The stop caps the loss when it does.

The Full Pairs Recipe

Everything Week 7 Unit 1

The whole week comes together here. Confirm the pair is cointegrated, estimate the hedge ratio, and form the spread. Read the adjustment speed for the half-life. Standardize into a z-score, then fade the extremes, exit at the mean, and stop if the equilibrium breaks. Cointegration is the foundation; error

The reversion (Case A)

- Alpha of -0.25 closes 25% a day.
- Half-life of 2.4 days.
- A fast, tradable spread.

The trade (Case B)

- Enter at z of +2, spread 6.44.
- Exit at z of 0, capturing 1.24.
- Stop at z of +3.

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Reproduce the Reversion and the Trade

Direct Claude Code to check your arithmetic

GOAL Compute the half-life, then backtest the z-score rules on a cointegrated spread.

[INPUT] Prompt · Claude Code

Write and run `part3_practice.py` that, on a cointegrated spread (reuse Part 1/2's, mean 5.2, sd 0.62):

CASE A - reversion:

- * fit the spread's AR(1); print alpha, phi, and the half-life $\ln(0.5)/\ln(\phi)$.

CASE B - the trade:

- * compute the z-score;
- * enter at $|z|=2$, exit at 0, stop at $|z|=3$;
- * print the entry/exit spread levels and the captured points per round trip.

Then plot the decay and the z-score with the bands.

Check against the slides

- alpha -0.25, phi 0.75.
- half-life 2.4 days.
- entry 6.44, exit 5.20.
- captured 1.24 per trade.

Deliverable: `part3_practice.py` whose printed numbers match the half-life and the round-trip trade.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 What does a negative adjustment coefficient mean?

**A restoring force
pulling the spread
back**

Q2 How do you get the half-life from the AR(1) coefficient?

**Log of one-half over
log of phi**

Q3 When do you stop out of a pairs trade?

**When the z-score
keeps widening past
the stop**

All three come from the adjustment coefficient, the half-life, and the z-score rules.

Three Pitfalls to Avoid

Where pairs trading trips people

Common mistakes

- "A stretched spread must revert soon."
- "Hold through any widening - it always comes back."
- "The hedge ratio is fixed forever."

The reality

- A long half-life means it may take weeks.
- Cointegration can break; the stop is essential.
- Re-estimate beta as the relationship drifts.

Two Calculations to Remember

Part 3 applications in one screen

1

The adjustment speed sets the half-life

An error-correction coefficient of -0.25 closes 25% of the gap each day, an AR(1) coefficient of 0.75. The half-life - $\log$ of one-half over $\log$ of 0.75 - is 2.4 days, the expected holding period of a trade in this spread.

2

The z-score turns the spread into a trade

Standardizing the spread (mean 5.2, sd 0.62), a z of +2 is a spread of 6.44 - short it - and a z of 0 is 5.20, where you cover, capturing 1.24 points. A z of +3 at 7.06 is the stop, in case the equilibrium breaks.

3

Cointegration, error correction, the z-score

The pair's shared trend is the foundation, error correction is the engine that reverts the spread, and the z-score is the trigger that times entry, exit, and the stop. That is the full pairs recipe.

Week 7 complete — from the idea of cointegration, through testing it, to the error-correction model and a live pairs trade.